

RUBEN DARIO FLOREZ ZELA

Electronic Engineer · Human-Centered AI for Intelligent Mobility · Computer Vision · Trustworthy & Edge AI

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ORCID | [Google Scholar](#) | [LinkedIn](#) | [GitHub](#) | [Personal Web](#)

RENACYT Level V (CONCYTEC) · Google Scholar: 234 citations · h-index 5 · i10-index 3 (Oct. 2026)

RESEARCH PROFILE

Electronic Engineer and AI researcher working on human-centered artificial intelligence for intelligent mobility. My research focuses on computer vision, trustworthy AI, and deployable intelligent systems, with growing interest in multimodal perception and the integration of human, environmental, and operational information for safer intelligent vehicles. My recent work explores robust perception under domain shift, explainable AI, uncertainty and conformal risk control, human-aware decision support, and real-time deployment on embedded hardware.

EDUCATION

M.Sc. in Electronic Engineering (Automation & Instrumentation)

Academic requirements completed Jul. 2026

Universidad Nacional de San Agustín de Arequipa (UNSA), Arequipa, Peru

- **48 credits completed; weighted average: 17.10/20** (official passing grades: 13–20). Degree expected 2026.
- **M.Sc. thesis:** *Multi-Horizon Prediction of Delayed-Response Risk and Three-State Conformal Supervision for Longitudinal Fallback in SAE Level 2 Automated Driving.*

Bachelor's Degree in Electronic Engineering

Universidad Nacional de San Antonio Abad del Cusco (UNSAAC), Cusco, Peru

Degree awarded Feb. 2023

- 223 approved academic credits; relevant coursework includes Artificial Intelligence, Robotics, Digital Signal Processing, Control Theory, Industrial Control Systems, Instrumentation, and Microprocessors and Computer Architecture.

Professional Title: Electronic Engineer

UNSAAC, Cusco, Peru | Professional License (CIP): 336757 | Obtained by thesis

Awarded Feb. 2024

- **Professional thesis:** *Design and Implementation of a Real-Time Driver Drowsiness Detection System Using Computer Vision and Convolutional Neural Networks; unanimously approved with excellence (19/20).*

SELECTED RESEARCH EXPERIENCE

Safety-Critical Risk Control for Automated Driving (M.Sc. Research)

2025 – 2026

Universidad Nacional de San Agustín de Arequipa (UNSA)

- Developed driver-grouped machine-learning models for multi-horizon prediction of delayed-response risk in SAE Level 2 automated driving.
- Designed calibrated and conformal risk-control mechanisms for graded longitudinal intervention with HUMAN–ASSIST–FALLBACK supervision.
- Evaluated safety effects through counterfactual longitudinal simulation under driver-specific uncertainty.

Explainable AI Auditing for Safety-Critical Perception (Lead Researcher)

2026

Independent research

- Developed a confidence-controlled framework for auditing pedestrian-detector explanations under cross-domain distribution shift.
- Evaluated explanation faithfulness using detector-specific perturbation methods while controlling detector confidence as a potential confounding factor.
- Resulting work accepted as a Regular Paper at **IEEE ICVES 2026**.

Deployment-Aware Computer Vision under Domain Shift (Lead Researcher)

2026 – Present

Independent research / public reproducibility repository

- Benchmarked lightweight CNN and transformer-based vision models for eye-state recognition under clean, corrupted, and cross-domain conditions.
- Evaluated robustness and embedded deployment on NVIDIA Jetson Nano/TensorRT using latency, throughput, memory, and reliability criteria.
- Developed reproducible and audit-oriented evaluation protocols for deployment-sensitive computer vision systems.

Embedded Vision for Driver-State Monitoring (Lead Researcher)

2023

LIECAR Lab, UNSAAC

- Developed CNN-based real-time driver monitoring using eye and facial-region analysis and deployed real-time inference on NVIDIA Jetson Nano.
- Designed ROI correction procedures robust to illumination variability.

SELECTED RESEARCH OUTPUTS

Accepted peer-reviewed conference paper

- [1] **Florez-Zela, R. D.** *Confidence-Controlled XAI Auditing for Pedestrian Detection under Domain Shift*. Accepted as a Regular Paper for the *2026 IEEE International Conference on Vehicular Electronics and Safety (ICVES)*.

Current journal manuscript and preprint

- [1] **Florez-Zela, R. D.** *Driver-Grouped Conformal Risk Control for Graded Longitudinal Intervention in Level 2 Automated Driving: A Counterfactual Simulation Study*. Manuscript submitted to *IEEE Transactions on Intelligent Vehicles*, 2026.
- [2] **Florez-Zela, R. D.** *Beyond Aggregate Scores: Deployment-Aware and Non-Compensatory Benchmarking of Vision-Based Eye-State Recognition Models for Driver Monitoring*. arXiv preprint 2606.08123, 2026.

Peer-reviewed journal articles

- [1] **Florez, R.**, Palomino-Quispe, F., Alvarez, A. B., Coaquira-Castillo, R. J., & Herrera-Levano, J. C. (2024). A Real-Time Embedded System for Driver Drowsiness Detection Based on Visual Analysis of the Eyes and Mouth Using Convolutional Neural Network and Mouth Aspect Ratio. *Sensors*, 24(19), 6261.
- [2] **Florez, R.**, Palomino-Quispe, F., Coaquira-Castillo, R. J., Herrera-Levano, J. C., Paixão, T., & Alvarez, A. B. (2023). A CNN-Based Approach for Driver Drowsiness Detection by Real-Time Eye State Identification. *Applied Sciences*, 13(13), 7849.

Selected peer-reviewed conference paper

- [1] Paixão, T., Alvarez, A. B., **Florez, R.**, & Palomino-Quispe, F. (2023). Fuzzy Control for Simplified Lumbar Spine Robotic Mechanism Motion. *2023 IEEE International Conference on Networking, Sensing and Control (ICNSC)*, pp. 1–6. IEEE.

Full publication list: [Google Scholar](#)

ACADEMIC POSITIONS

Contract Lecturer – Electronic Engineering Department

Aug. 2024 – Jan. 2026

Universidad Nacional de San Antonio Abad del Cusco (UNSAAC), Cusco, Peru

- Taught Artificial Intelligence, Digital Image Processing, Robotics, and Electronics Laboratories; supervised applied student work in machine learning, computer vision, and embedded systems.

Teaching Assistant – Electronic Engineering Department

Apr. 2024 – Aug. 2024

Universidad Nacional de San Antonio Abad del Cusco (UNSAAC), Cusco, Peru

- Supported practical and laboratory sessions in electronics and AI-related courses.

TECHNICAL SKILLS

Programming: Python, C/C++, MATLAB

Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn; classification, calibration, uncertainty-aware evaluation

Computer Vision & XAI: OpenCV, CNN/transformer-based vision models, pedestrian detection, RGB/NIR imaging, visual explanation auditing, gaze and facial analysis

Embedded & Edge AI: NVIDIA Jetson Nano, TensorRT, CUDA-enabled inference, Raspberry Pi, STM32, ESP32

Control & Real-Time: Digital control, PID/fuzzy control, real-time automation, instrumentation

Research Software: Linux, Git/GitHub, reproducible experiment organization, deterministic evaluation workflows

RESEARCH INTERESTS

· Human-Centered AI · Intelligent Mobility · Computer Vision · Multimodal Perception · Human-Aware Intelligent Vehicles · Trustworthy & Explainable AI · Uncertainty & Risk-Aware AI · Embedded & Edge AI

SELECTED DISTINCTIONS, SERVICE & LANGUAGES

- **RENACYT Level V Researcher**, CONCYTEC, Peru; IEEE Member (ITSS and VTS).
- **1st Place:** VIII Regional Science, Technology & Innovation Fair (2023) and II International Congress of Research, Innovation & Entrepreneurship, UNSAAC (2022).
- Reviewer for *Scientific Reports*, *Scientific Data*, and *ACM Transactions on Intelligent Systems and Technology*.
- **Languages:** Spanish (native); English (working proficiency for academic research and scientific communication).